

Climate 5000 Series - Gen 4

12K BTU Single Zone Max Performance System

Wall Mounted IDU

Engineering
Submittal
Sheet



BOSCH

Overview and Project Information



Project name

Location

Architect

Engineer

General contractor

Reference #

12K Btu Single Zone Max Performance System - Wall Mounted IDU

Indoor part number		8-733-962-673
Indoor model number		BMS500-AAU012-1AHWXD
Outdoor part number		8-733-962-689
Outdoor model number		BMS500-AAS012-1CSXHD
Efficiency Data		AHRI 210/240 2023
SEER2		25.4
EER2		13.4
HSPF2 Region IV		10.5
HSPF2 Region V		8.0
COP(47F)		3.76
COP (5F)		1.80
Performance Data		AHRI 210/240 2023
Cooling	Capacity Range	3800~14600
	Rated Capacity	12000
	Current	4.0
	Input Power	890
Heating	Capacity Range	3800~16000
	Rated Capacity (47F)	12000
	Rated Capacity (5F)	11500
	Current	4.1
	Input Power	935
Electrical Data		
Power supply (V / Ø / Hz)		208-230 / 1 / 60
Min. circuit ampacity indoor/outdoor (A)		3.0 / 15.0
Max. fuse indoor/outdoor (A)		15.0 / 15.0
Compressor Data		
Type:		Rotary
Refrigerant oil /oil charge (ml)		VG74 / 350
Input (W)		885
Rated current (RLA/A)		6.0
Operation Range		
ODU	Cooling (°F / °C)	-22~122 / -30~50
	Heating (°F / °C)	-22~75 / -30~24
IDU	Cooling (°F / °C)	60~90 / 16~32
	Heating (°F / °C)	32~86 / 0~30

Piping

Refrigerant precharge (oz/g)	35.27 / 1000
Length for precharge (ft/m)	24.6 / 7.5
Liquid side (in/mm)	1/4" / Ø6.35
Gas side (in/mm)	3/8" / Ø9.52
Additional Charge of Refrigerant (oz/ft)/(g/m)	0.16 / 15
Max. refrigerant pipe length (ft/m)	82 / 25
Max. difference in height (ft/m)	49.2 / 15

System Data

Refrigerant type		R454B
Design pressure (PSIG)		550 / 340
Dehumidification (pts/hr)		2.53
IDU	Fan motor model	ZKFP-20-8-6-7
	Fan motor qty (ea)	1
	Fan motor input (W)	50.0
	Fan speed (RPM)	1100 / 850 / 700
	Air flow, Turbo / H / M / L / Si (CFM)	441.4 / 323.7 / 229.6 / 176.6 / 147.2
	Approx. Air Throw Range ft (m)	23.0 / 7
	Noise level, Turbo / H / M / L / Si (dB(A))	47 / 36.5 / 32.5 / 22 / 19.5
	Net weight (lbs / kg)	22.93 / 10.4
	Gross weight (lbs / kg)	29.32 / 13.3
ODU	Fan motor model	ZKFN-34-10-1-3
	Fan motor qty (ea)	1
	Fan motor input (W)	50.0
	Fan motor speed, H / / L (RPM)	780 / 680 / 580
	Air flow, Max (CFM)	1470.6
	Noise level, Max (dB(A))	56
	Net weight (lbs / kg)	72.75 / 33
	Gross weight (lbs / kg)	78.7 / 35.7

Accessories

BMS-WT2-XXD (8733962707)	Wired Wall Thermostat
BMS-DRY-XXD (8733969263)	24V Interface
BMS-PK2-7X1 (8733951011)	25ft Lineset

Standard Parts

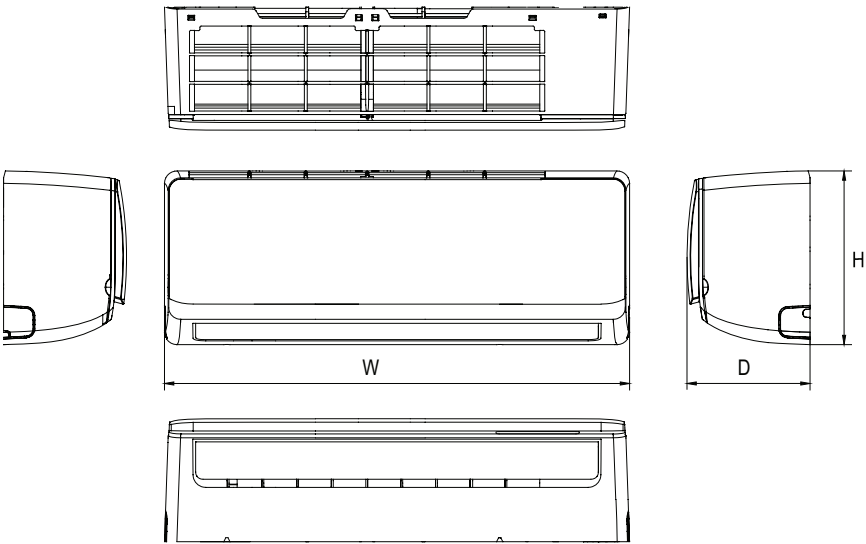
8-733-969-088	Remote Control
---------------	----------------

Climate 5000 Series - Gen 4

12K Btu Single Zone Max Performance System - Wall Mounted IDU

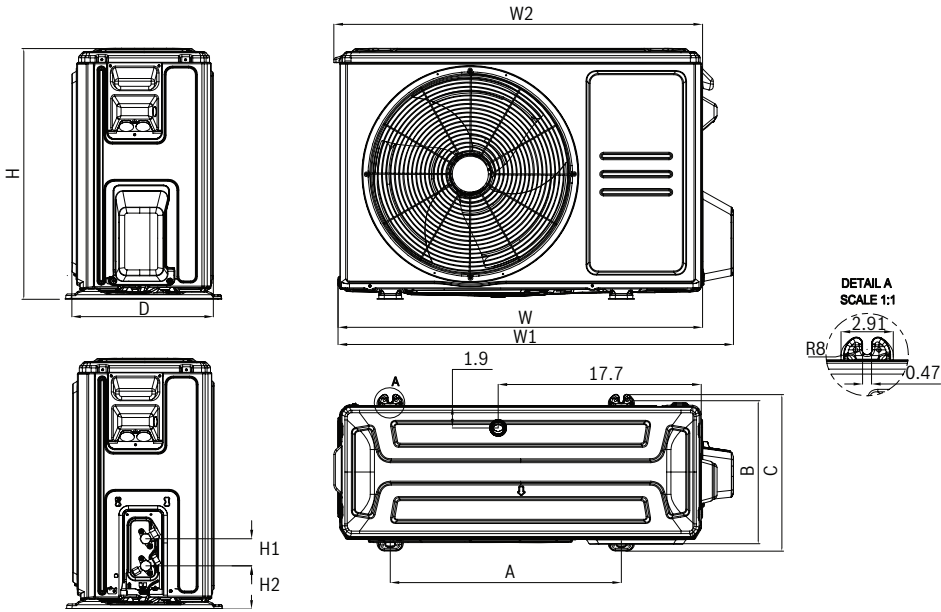


Indoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)
BMS500-AAU012-1AHWXD	31.42	9.21	11.85

Outdoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)	W1 (in)	W2 (in)	H1 (in)	H2 (in)	A (in)	B (in)	C (in)
BMS500-AAS012-1CSXHD	31.69	12.99	21.81	34.4	32.1	2.4	3.8	20.1	12.5	13.6

Climate 5000 Series - Gen 4

12K Btu Single Zone Max Performance System - Wall Mounted IDU


BOSCH

Cooling Performance - Indoor Dry Bulb Temperature at 441 CFM																					
ODU DB(°F)	IDU WB(°F)	60.8°F(16.0°C)					64.4°F(18.0°C)					67°F(19.4°C)					71.6°F(22.0°C)				
	IDU DB(°F)	73.4°F(23.0°C)	77.0°F(25.0°C)	80.0°F(26.7°C)	80.6°F(27.0°C)	84.2°F(29.0°C)	73.4°F(23.0°C)	77.0°F(25.0°C)	80.0°F(26.7°C)	80.6°F(27.0°C)	84.2°F(29.0°C)	73.4°F(23.0°C)	77.0°F(25.0°C)	80.0°F(26.7°C)	80.6°F(27.0°C)	84.2°F(29.0°C)	73.4°F(23.0°C)	77.0°F(25.0°C)	80.0°F(26.7°C)	80.6°F(27.0°C)	84.2°F(29.0°C)
-22°F (-30.0°C)	TC	9549	9545	9681	9721	9989	9963	9955	9949	9948	9940	10261	10248	10239	10237	10228	10824	10821	10799	10795	10772
	S/T	0.80	0.91	0.99	0.99	0.99	0.64	0.75	0.84	0.86	0.97	0.55	0.65	0.74	0.75	0.86	0.39	0.47	0.55	0.57	0.67
	PI	0.58	0.58	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.58	0.58	0.58	0.58	0.58
-13°F (-25.0°C)	TC	10345	10340	10488	10531	10821	10794	10785	10778	10777	10768	11116	11102	11092	11090	11080	11726	11723	11698	11694	11669
	S/T	0.80	0.91	1.00	1.00	1.00	0.65	0.75	0.85	0.86	0.97	0.55	0.65	0.74	0.76	0.86	0.39	0.47	0.56	0.57	0.67
	PI	0.60	0.60	0.60	0.60	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.60	0.60	0.60	0.60	0.60
-4°F (-20.0°C)	TC	11141	11136	11295	11342	11654	11624	11614	11607	11606	11596	11972	11957	11945	11944	11932	12628	12624	12598	12594	12567
	S/T	0.80	0.92	1.00	1.00	1.00	0.65	0.76	0.85	0.87	0.98	0.55	0.65	0.74	0.76	0.86	0.39	0.48	0.56	0.57	0.67
	PI	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.61	0.61	0.61	0.61	0.62	0.62	0.62	0.62	0.62	0.62
0°F (-17.8°C)	TC	11511	11504	11717	11766	12089	12004	11991	11982	11980	12101	12357	12340	12324	12323	12317	13022	13019	13018	13019	13028
	S/T	0.81	0.93	1.00	1.00	1.00	0.66	0.77	0.86	0.88	1.00	0.55	0.66	0.75	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.63	0.63	0.63	0.63	0.63
5°F (-15.0°C)	TC	11841	11882	12153	12209	12585	12396	12399	12400	12413	12598	12792	12796	12798	12799	12816	13543	13539	13546	13547	13553
	S/T	0.82	0.94	1.00	1.00	1.00	0.66	0.78	0.87	0.88	1.00	0.56	0.67	0.76	0.78	0.88	0.39	0.48	0.57	0.59	0.69
	PI	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.64	0.64	0.64	0.64	0.64	0.65	0.65	0.65	0.65	0.65
14°F (-10.0°C)	TC	12714	12707	12967	13028	13432	13345	13333	13324	13323	13447	13802	13785	13771	13769	13755	14676	14672	14643	14638	14607
	S/T	0.81	0.93	1.00	1.00	1.00	0.65	0.76	0.86	0.87	1.00	0.55	0.66	0.75	0.76	0.87	0.39	0.48	0.56	0.57	0.67
	PI	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.59	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60
17°F (-8.3°C)	TC	12973	13016	13317	13382	13816	13619	13625	13627	13639	13831	14084	14089	14095	14096	14113	14979	14975	14972	14975	14988
	S/T	0.82	0.93	1.00	1.00	1.00	0.66	0.77	0.87	0.88	1.00	0.55	0.66	0.76	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.59	0.59	0.59	0.59	0.59
23°F (-5.0°C)	TC	12379	12432	12754	12817	13239	13006	13010	13011	13028	13253	13455	13459	13463	13463	13485	14306	14302	14312	14314	14322
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78	0.89	0.39	0.48	0.57	0.59	0.69
	PI	0.60	0.60	0.60	0.60	0.61	0.60	0.60	0.60	0.60	0.61	0.61	0.61	0.61	0.61	0.61	0.62	0.62	0.62	0.62	0.62
35°F (1.7°C)	TC	11300	11300	11366	11398	11766	11828	11827	11828	11828	11860	12198	12197	12198	12198	12199	12885	12882	12881	12881	12882
	S/T	0.78	0.88	0.97	0.99	1.00	0.63	0.73	0.82	0.83	0.93	0.54	0.63	0.72	0.73	0.83	0.39	0.46	0.54	0.56	0.65
	PI	0.60	0.60	0.60	0.60	0.63	0.63	0.63	0.63	0.63	0.64	0.66	0.66	0.66	0.66	0.66	0.72	0.72	0.72	0.72	0.72
41°F (5.0°C)	TC	11267	11267	11357	11436	11774	11791	11791	11791	11833	12159	12158	12159	12159	12159	12160	12841	12838	12837	12836	12838
	S/T	0.78	0.89	0.98	1.00	1.00	0.64	0.74	0.83	0.84	0.94	0.54	0.64	0.72	0.74	0.84	0.39	0.47	0.55	0.56	0.66
	PI	0.61	0.61	0.61	0.62	0.64	0.65	0.65	0.65	0.65	0.65	0.67	0.67	0.68	0.68	0.68	0.74	0.74	0.73	0.73	0.74
47°F (8.3°C)	TC	11234	11241	11392	11443	11781	11755	11754	11755	11754	11809	12120	12119	12119	12119	12119	12798	12795	12794	12793	12795
	S/T	0.79	0.90	1.00	1.00	1.00	0.64	0.75	0.84	0.85	0.96	0.54	0.65	0.73	0.75	0.85	0.39	0.47	0.55	0.57	0.67
	PI	0.62	0.62	0.63	0.63	0.66	0.66	0.66	0.66	0.66	0.66	0.69	0.69	0.69	0.69	0.69	0.75	0.75	0.75	0.75	0.75
50°F (10.0°C)	TC	11141	11147	11834	11902	12364	11770	11776	11857	11895	12375	12229	12245	12251	12252	12370	13072	13106	13140	13144	13160
	S/T	0.90	1.00	1.00	1.00	1.00	0.71	0.85	0.96	0.99	1.00	0.59	0.72	0.83	0.85	0.99	0.40	0.50	0.60	0.62	0.74
	PI	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69
59°F (15.0°C)	TC	11463	11865	12286	12361	12864	12128	12145	12283	12374	12877	12629	12642	12646	12664	12887	13561	13588	13615	13618	13629
	S/T	0.92	1.00	1.00	1.00	1.00	0.73	0.86	1.00	1.00	1.00	0.60	0.73	0.85	0.86	1.00	0.40	0.51	0.61	0.63	0.76
	PI	0.68	0.68	0.68	0.68	0.69	0.68	0.68	0.68	0.68	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69
68°F (20.0°C)	TC	11778	12280	12736	12817	13361	12469	12497	12750	12831	13374	13008	13017	13041	13052	13385	14016	14042	14064	14066	14073
	S/T	0.95	1.00	1.00	1.00	1.00	0.74	0.89	1.00	1.00	1.00	0.61	0.75	0.87	0.89	1.00	0.40	0.51	0.63	0.65	0.78
	PI	0.67	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.69	0.69	0.69	0.69	0.69
77°F (25.0°C)	TC	12080	12662	13157	13245	13837	12806	12847	13172	13260	13851	13386	13401	13436	13451	13862	14458	14502	14535	14539	14552
	S/T	0.97	1.00	1.00	1.00	1.00	0.75	0.90	1.00	1.00	1.00	0.61	0.76	0.88	0.90	1.00	0.40	0.52	0.63	0.65	0.79
	PI	0.67	0.67	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.69	0.69	0.69	0.69	0.69
82°F (27.8°C)	TC	12127	12746	13260	13350	13961	12851	12900	13272	13365	13976	13448	13465	13507	13524	13988	14541	14595	14633	14637	14653
	S/T	0.99	1.00	1.00	1.00	1.00	0.76	0.91	1.00	1.00	1.00	0.62	0.77	0.89	0.92	1.00	0.40	0.52	0.64	0.66	0.80
	PI	0.73	0.73	0.74	0.74	0.74	0.73	0.73	0.74	0.74	0.74	0.74	0.74	0.74	0.74	0.74	0.75	0.75	0.75	0.75	0.74
86°F (30.0°C)	TC	12054	12655	13173	13265	13883	12712	12770	13188	13280	13897	13314	13329	13378	13397	13909	14414	14470	14504	14507	14521
	S/T	1.00	1.00	1.00	1.00	1.00	0.77	0.93	1.00	1.00	1.00	0.62	0.78	0.91	0.93	1.00	0.40	0.52	0.64	0.66	0.81
	PI	0.85	0.86	0.86	0.86	0.87	0.86	0.86	0.86	0.86	0.87	0.86	0.86	0.86	0.86	0.87	0.87	0.87	0.87	0.87	0.87
95°F (35.0°C)	TC	1171																			

Climate 5000 Series - Gen 4

12K Btu Single Zone Max Performance System - Wall Mounted IDU



Heating Performance - Indoor Dry Bulb Temperature at 441 CFM										
Outdoor DB(°F)	TC: Total Heating Capacity(BTU/hr)					PI: Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-22°F(-30.0°C)	6438	6310	6278	6253	6197	1.43	1.45	1.45	1.45	1.46
-13°F(-25.0°C)	7768	7613	7575	7544	7476	1.55	1.56	1.56	1.57	1.57
-4°F(-20.0°C)	9497	9307	9260	9223	9140	1.68	1.69	1.70	1.70	1.71
0°F(-17.8°C)	10450	10241	10190	10149	10058	1.69	1.71	1.71	1.71	1.72
5°F(-15.0°C)	11642	11409	11352	11306	11205	1.71	1.72	1.73	1.73	1.74
14°F(-10.0°C)	12836	12579	12516	12466	12354	2.03	2.05	2.05	2.05	2.06
17°F(-8.3°C)	13234	12969	12904	12852	12737	2.14	2.15	2.16	2.16	2.17
22°F(-5.6°C)	13893	13615	13547	13493	13371	2.12	2.14	2.14	2.15	2.16
27°F(-2.8°C)	14552	14261	14190	14133	14006	2.11	2.12	2.13	2.13	2.14
32°F(0.0°C)	15211	14907	14833	14773	14640	2.09	2.11	2.11	2.12	2.12
35°F(1.7°C)	15607	15295	15218	15157	15021	2.08	2.10	2.10	2.11	2.11
37°F(2.8°C)	15754	15439	15361	15300	15162	1.97	1.98	1.99	1.99	2.00
42°F(5.6°C)	16121	15798	15719	15657	15516	1.68	1.69	1.70	1.70	1.71
44.6°F(7.0°C)	16268	15942	15863	15799	15657	1.57	1.58	1.58	1.58	1.59
47°F(8.3°C)	16488	16158	16077	16013	15869	1.39	1.41	1.41	1.41	1.42
52°F(11.1°C)	15812	15496	15419	15357	15219	1.25	1.26	1.26	1.26	1.27
57°F(13.9°C)	15137	14834	14760	14701	14569	1.10	1.11	1.11	1.12	1.12
59°F(15.0°C)	14867	14569	14497	14439	14309	1.04	1.05	1.05	1.06	1.06
62°F(16.7°C)	14461	14172	14101	14045	13919	0.96	0.96	0.97	0.97	0.97
64.4°F(18.0°C)	14137	13854	13785	13730	13606	0.89	0.89	0.90	0.90	0.90
69.8°F(21.0°C)	13408	13139	13074	13022	12904	0.73	0.74	0.74	0.74	0.74
75.2°F(24.0°C)	12678	12425	12362	12313	12202	0.57	0.58	0.58	0.58	0.58

Bolded values were tested. All other values are simulated and for reference only.